

ZIMSEC MS 6042/2

MR SHARE  ONLINE

REVISION .ft

[TUKS. t: 1~2 testing]

General Certificate of Education Advanced Level

PURE MATHEMATICS

6042/2

PAPER 2

SOLUTION

JUNE 2024 SESSION

3 hours

Additional materials:

Answer paper

Graph paper

List of formulae MF7

Electronic calculator (Non-programmable)

Time 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name in the spaces provided on the answer sheet/answer booklet.

Answer **all** questions in **section A** and any **five** from **section B**.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given to 2 significant figures.

INFORMATION TO CANDIDATES

The number of marks is given in brackets [] at the end of each question or part of question.

The total number of marks for this paper is 120.

The use of a scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 7 printed pages and 1 blank page.

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Section A (40 marks)

Answer **all** question in this section.

1. The extension y , of an elastic string varies directly as the magnitude of force F , exerted in extending it. If $F = 6$ Newtons and $y = 0.45m$

(a) Express F as a function of y . [4]

(b) Hence or otherwise find F when $y = 0.2m$. [2]

SOLUTION

(a) $y \propto F$

$$y = kF$$

$$\text{When } F = 6, y = 0.45$$

$$0.45 = k6$$

$$k = \frac{3}{40}$$

$$y = \frac{3}{40}F$$

$$\therefore F = \frac{40}{3}y$$

(b) When $y = 0.2$

$$F = \frac{40}{3}(0.2) = \frac{8}{3} = 2\frac{2}{3} \text{ Newtons}$$

2. Set S contains functions $g(x)$, $f(x)$ and $h(x)$ which are defined by

$$g: x \rightarrow x$$

$$f: x \rightarrow x - 1$$

$$h: x \rightarrow x + 1$$

$\circ$	x	$x - 1$	$x + 1$
x	x	$x - 1$	$x + 1$
$x - 1$			x
$x + 1$	$x + 1$		

(a) Copy and complete this table. [2]

(b) Use the table to state,

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- (i) the identity element of the set, [1]
 (ii) the inverse of each element of the set, [3]
 (iii) why the set S is not a group under function composition. [2]

SOLUTION

(a)

$\circ$	x	$x - 1$	$x + 1$
x	x	$x - 1$	$x + 1$
$x - 1$	$x - 1$	$x - 2$	x
$x + 1$	$x + 1$	x	$x + 2$

(b)

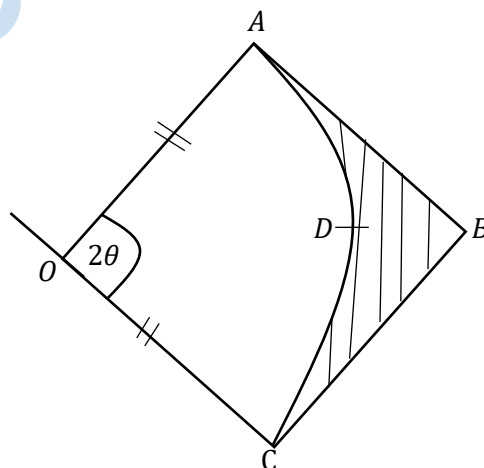
(i) the identity element, $e = x$

(ii) the inverse of;

 x is x or/ $g(x)$ is $g(x)$ [self inverse] $x - 1$ is $x + 1$ or/ $f(x)$ is $h(x)$ $x + 1$ is $x - 1$ or/ $h(x)$ is $f(x)$

- iii. Since there are **new functions** in the Cayley table, **Closure property is not upheld i.e $\{(x - 2), (x + 2)\}$** , hence the set S **does not form a group** under composition of functions.

3.



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In the diagram OABC is a rhombus. ADC is an arc of a circle centre O, radius r and subtends an angle 2θ at O. the shaded region is bounded by the arc ADC and sides AB and CB.

(a) Show that the area of the shaded region is $r^2(\sin 2\theta - \theta)$. [6]

(b) Given that $r = 8\text{cm}$ and $\theta = \frac{\pi}{6}$ radians, find the exact area of the shaded region. [2]

SOLUTION

(a) Shaded area = Area of rhombus OABC – Area of sector OADC

$$\begin{aligned}
 &= r^2 \sin \beta - \frac{1}{2} r^2 \beta \\
 &= r^2 \sin(2\theta) - \frac{1}{2} r^2 (2\theta) \quad [\beta = 2\theta] \\
 &= r^2 \sin(2\theta) - r^2 \theta \\
 &= r^2 (\sin 2\theta - \theta) \quad [\text{shown}]
 \end{aligned}$$

(b) Given that $r = 8\text{cm}$ and $\theta = \frac{\pi}{6}$ radians, shaded area is;

$$\begin{aligned}
 &= 8^2 \left[\sin\left(2 \cdot \frac{\pi}{6}\right) - \frac{\pi}{6} \right] \\
 &= 64 \left[\sin \frac{\pi}{3} - \frac{\pi}{6} \right] \\
 &= \left(32\sqrt{3} - \frac{32}{3}\pi \right) \text{cm}^2
 \end{aligned}$$

4. An experiment is performed to find corresponding values of x and y . Results were recorded in the table below.

x	1	2	3	4	5	6
y	-1	2	9	20	35	54

It is suggested that the variables x and y are connected by the equation $y = px^2 + qx$ where p and q are constants.

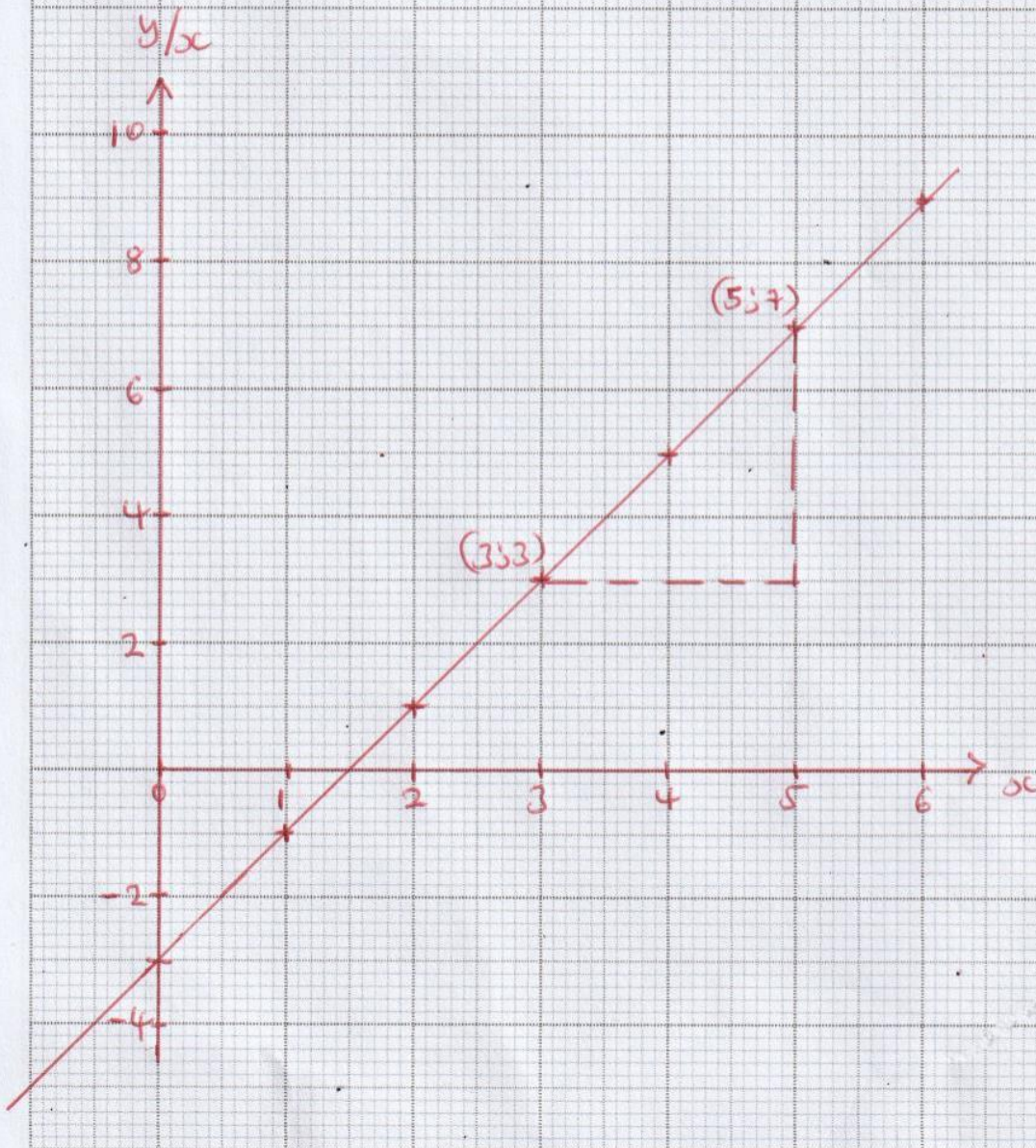
(a) By plotting $\frac{y}{x}$ against x , draw graph for the given data. [5]

(b) Use the graph to find the values of p and q . [4]

SOLUTION

Question: Q 49)

x	1	2	3	4	5	6
$\frac{y}{x}$	-1	1	3	5	7	9



$$(b) \frac{y}{x} = p \frac{x^2}{x} + \frac{qx}{x}$$

$$\frac{y}{x} = px + q$$

From the graph;

$$q = -3$$

$$p = m = \frac{7-3}{5-3} = \frac{4}{2} = 2$$

5.

$$(a) \text{ Show that } x^3 + x^2 + 2x + 2 \equiv (x+1)(x^2+2) \quad [2]$$

$$(b) \text{ Hence or otherwise express } \frac{x^3}{x^3+x^2+2x+2} \text{ in partial fractions.} \quad [7]$$

SOLUTION

$$(a) x^3 + x^2 + 2x + 2 \equiv (x+1)(x^2+2)$$

$$\text{RHS} = x(x^2+2) + 1(x^2+2)$$

$$= x^3 + 2x + x^2 + 2$$

$$= x^3 + x^2 + 2x + 2 \equiv \text{LHS [shown]}$$

(b)

$$\begin{array}{r|l} 1 & \\ \hline x^3 + x^2 + 2x + 2 & x^3 \\ & -(x^3 + x^2 + 2x + 2) \\ \hline & -x^2 - 2x - 2 \end{array}$$

$$\frac{x^3}{x^3 + x^2 + 2x + 2} = A + \frac{B}{x+1} + \frac{Cx+D}{x^2+2}$$

$$A = 1$$

When $x = -1$;

$$B = \frac{-1 + 2 - 2}{1 + 2} = -\frac{1}{3}$$

When $x = 0$;

$$-2 = -\frac{1}{3}(2) + D(1)$$

$$D = -\frac{4}{3}$$

When $x = 1$;

$$-1 - 2 - 2 = -\frac{1}{3}(3) + \left(C - \frac{4}{3}\right)(2)$$

$$-5 = -1 + 2C - \frac{8}{3}$$

$$-\frac{4}{3} = 2C$$

$$C = -\frac{2}{3}$$

$$\begin{aligned} \frac{x^3}{x^3 + x^2 + 2x + 2} &= 1 - \frac{1}{3(x+1)} + \frac{(-2x-4)}{3(x^2+2)} \\ &= 1 - \frac{1}{3(x+1)} - \frac{2x+4}{3(x^2+2)} \end{aligned}$$

Section B (80 marks)

Answer any **five** questions from this section. Each question carries **16** marks.

6. Matrix M maps A(2;5) and B(-1;4) onto $A^1(9; 15)$ and $B^1(-6; -9)$ respectively.

(a) Find Matrix M

(b) Point $C^1(-6; 15)$ is the image of point C under the transformation matrix M.

(i) Sketch Point A^1, B^1 and C^1 on a cartesian plane. [1]

(ii) Hence calculate area of triangle $A^1B^1C^1$. [2]

(c) Find the

(i) area of triangle ABC [4]

(ii) co-ordinate of C. [3]

SOLUTION

(a) let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 5 & -4 \end{pmatrix} = \begin{pmatrix} 9 & -6 \\ 15 & -9 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 9 & -6 \\ 15 & -9 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 5 & -4 \end{pmatrix}^{-1}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 9 & -6 \\ 15 & -9 \end{pmatrix} \cdot -\frac{1}{3} \begin{pmatrix} -4 & 1 \\ -5 & 2 \end{pmatrix}$$

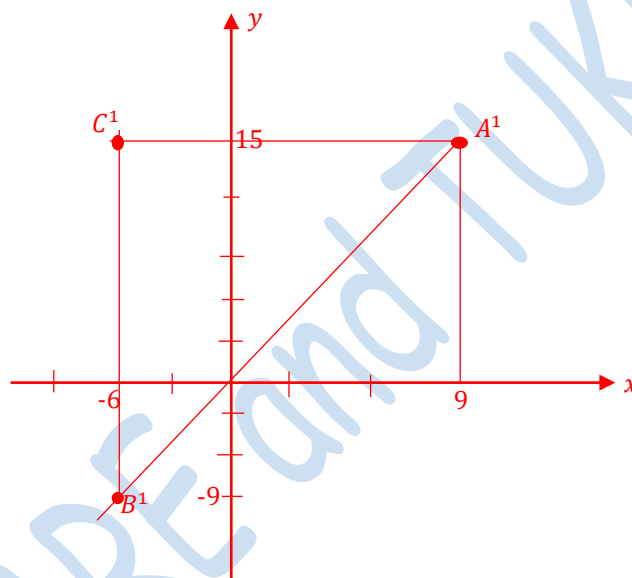
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} 9 & -6 \\ 15 & -9 \end{pmatrix} \begin{pmatrix} -4 & 1 \\ -5 & 2 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} -6 & -3 \\ -15 & -3 \end{pmatrix}$$

$$\therefore M = \begin{pmatrix} 2 & 1 \\ 5 & 1 \end{pmatrix}$$

(b)

(i) S



$$\begin{aligned} \text{(ii) Area of } \Delta A^1B^1C^1 &= \frac{1}{2} \times 15 \times 24 \\ &= \mathbf{180 \text{ units}} \end{aligned}$$

(c)

$$\text{(i) Area of } \Delta A^1B^1C^1 = |M| \times \text{Area of } \Delta ABC$$

$$180 = |2 - 5| \times \text{Area of } \Delta ABC$$

$$180 = 3 \times \text{Area of } \Delta ABC$$

$$\therefore \text{Area of } \Delta ABC = \mathbf{60 \text{ units}^2}$$

$$\text{(ii) } \begin{pmatrix} 2 & 1 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -6 \\ 15 \end{pmatrix}$$

$$2x + y = -6$$

$$y = -6 - 2x \dots \dots (1)$$

$$5x + y = 15 \dots \dots (2)$$

Solving equation (1) and (2) yields;

$$x = 7, y = -20$$

$$\therefore C(7, -20)$$

7. Given that a complex number W is such that $W(2 + 3i) = 9 - 6i$

(a) Express W in the form $x + iy$

(b) Hence or otherwise state

(i) $|W|$

(ii) $\arg W$.

[2]

(c) Using DeMoivre's theorem or otherwise find W^3 in trigonometric form.

[4]

(d) On an Argand diagram, shade the region which satisfy the loci,

$$|Z - W| \leq 3 \text{ and } -\frac{2\pi}{3} < \arg Z \leq -\frac{\pi}{3} \text{ simultaneously.}$$

[6]

SOLUTION

(a) $W(2 + 3i) = 9 - 6i$

$$W = \frac{9 - 6i}{2 + 3i} \times \frac{2 - 3i}{2 - 3i}$$

$$W = \frac{18 - 27i - 12i - 18}{4 + 9}$$

$$W = \frac{0}{13} - \frac{39}{13}i$$

$$W = 0 - 3i$$

(b)

(i) $|W| = \sqrt{0^2 + (-3)^2} = 3$

(ii) $\arg(W) = -\frac{\pi}{2}$

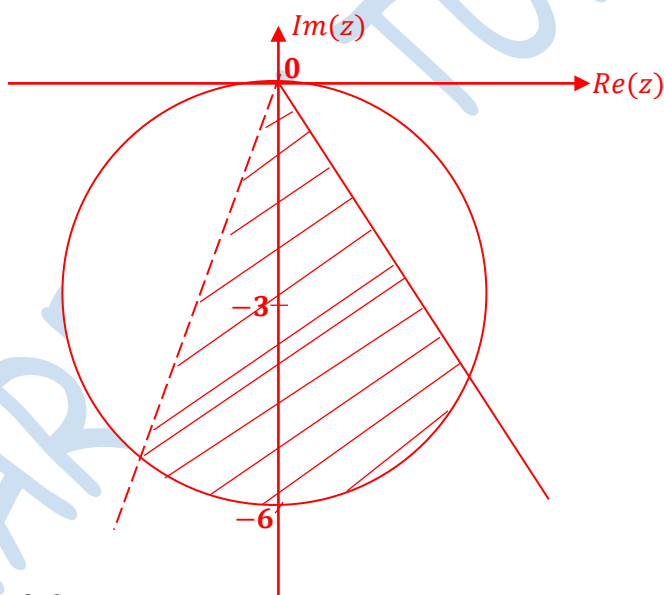
(c) $r = |W^3| = 3^3 = 27$

$$\begin{aligned} \theta &= \arg(W^3) \\ &= 3 \arg(W) \\ &= 3 \left(-\frac{1}{2} \pi \right) + 2\pi \\ &= \frac{1}{2} \pi \end{aligned}$$

$$\begin{aligned}\therefore W^3 &= r(\cos \theta + i \sin \theta) \\ &= 27 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right)\end{aligned}$$

(d)

$$\begin{aligned}|z - (0 - 3i)| &\leq 3 \\ -\frac{2}{3}\pi < \arg[z - (0 + 0i)] &\leq -\frac{\pi}{3}\end{aligned}$$



8.

(e) Express $\frac{1}{(r+1)(r+3)}$ in partial fractions.

[5]

(f) Hence or otherwise

(i) show that

$$\sum_{r=1}^n \frac{1}{(r+1)(r+3)} = \frac{5}{12} - \frac{2n+5}{2(n+2)(n+3)},$$

[6]

(ii) state the sum to infinity,

[1]

(iii) Evaluate

$$\sum_{r=12}^{25} \frac{1}{(r+1)(r+3)},$$

SOLUTION

$$(a) \frac{1}{(r+1)(r+3)} = \frac{A}{r+1} + \frac{B}{r+3}$$

$$\text{If } r = -1; \quad A = \frac{1}{2}$$

$$\text{If } r = -3; \quad B = -\frac{1}{2}$$

$$\frac{1}{(r+1)(r+3)} = \frac{1}{2(r+1)} - \frac{1}{2(r+3)}$$

(b)

(i)

$$\begin{aligned} & \sum_{r=1}^n \frac{1}{2} \left[\frac{1}{(r+1)} - \frac{1}{(r+3)} \right] \\ &= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} \right] \\ &= + \left[\frac{1}{3} - \frac{1}{5} \right] \\ &= + \left[\frac{1}{4} - \frac{1}{6} \right] \\ &\quad \cdot \\ &\quad \cdot \\ &= + \left[\frac{1}{n} - \frac{1}{n+2} \right] \\ &= + \left[\frac{1}{n+1} - \frac{1}{n+3} \right] \\ &= \frac{1}{2} \left[\frac{5}{6} - \left(\frac{n+3+n+2}{(n+2)(n+3)} \right) \right] \\ &= \frac{5}{12} - \frac{1}{2} \left(\frac{2n+5}{(n+2)(n+3)} \right) \\ &= \frac{5}{12} - \frac{2n+5}{2(n+2)(n+3)} \quad [\text{shown}] \end{aligned}$$

$$(ii) \quad S_{\infty} = \frac{5}{12}$$

(c)

$$\begin{aligned}
 \sum_{r=12}^{25} \frac{1}{(r+1)(r+3)} &= \left[\frac{5}{12} - \frac{2n+5}{2(n+2)(n+3)} \right]_{11}^{25} \\
 &= \frac{575}{1512} - \frac{187}{546} \\
 &= \frac{743}{19656} \\
 &= \mathbf{0.038 \text{ (2s.f)}}
 \end{aligned}$$

9. On this question there was an omission on the plane π_1 . Now given the lines m and l .

(a)

$$(i) \quad \vec{d}_m = \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} \text{ and } \vec{d}_l = \begin{pmatrix} -4 \\ 1 \\ 1 \end{pmatrix}$$

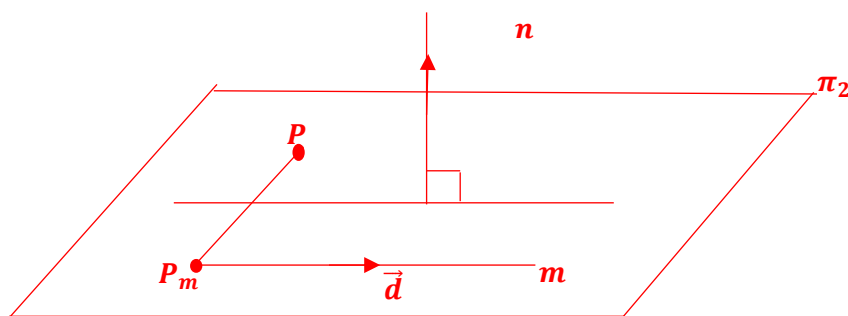
$$\begin{aligned}
 \vec{d}_m \cdot \vec{d}_l &= \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -4 \\ 1 \\ 1 \end{pmatrix} \\
 &= -4 - 1 + 5 \\
 &= \mathbf{0 \text{ [shown]}}
 \end{aligned}$$

(ii) Error on equation of plane (π_1), $r \cdot n = d$

$$\text{But; } \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix} = -9 - 2 - 2 = -13 \neq d \text{ [shown] } [d \text{ was omitted, (error)}]$$

(b)

(i)



$$\vec{d}_m = \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix}$$

$$\vec{d}_2 = \vec{OP} - \vec{OP}_m$$

$$= \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$$

$$n_{\pi_2} = \begin{vmatrix} i & j & k \\ 1 & -1 & 5 \\ -4 & -1 & 3 \end{vmatrix}$$

$$= i(-3 + 5) - j(3 + 20) + k(-1 - 4)$$

$$= 2i - 23k - 5k$$

$$\pi_2; \quad r \cdot \begin{pmatrix} 2 \\ -23 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ -23 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$$

$$r \cdot \begin{pmatrix} 2 \\ -23 \\ -5 \end{pmatrix} = -6 - 23 - 10$$

$$r \cdot \begin{pmatrix} 2 \\ -23 \\ -5 \end{pmatrix} = -39$$

$$(ii) \quad \pi_1; \quad 3x - 2y - z = d$$

$$\pi_2; \quad 2x - 23y - 5z = -39$$

Plane π_1 was incomplete (error)

But follow these procedures;

- Let $x = 0$ or $y = 0$ or $z = 0$ on both equations of planes to find the point 'a' on line of intersection of both planes.
- The for the direction of line, $\vec{d}$

$$\vec{d} = n_{\pi_1} \times n_{\pi_2}$$

$$= \begin{vmatrix} i & j & k \\ 3 & -2 & -1 \\ 2 & -23 & -5 \end{vmatrix}$$

$$= i(10 - 23) - j(-15 + 2) + k(-69 + 4)$$

$$= -13i + 13j - 65k$$

$$= \begin{pmatrix} -13 \\ 13 \\ -65 \end{pmatrix}$$

- Taking point on line as 'a'

Equation of line $r = a + \mu \cdot \vec{d}$

$$r = a + \mu \cdot \begin{pmatrix} -13 \\ 13 \\ -65 \end{pmatrix}$$

$$r = a + \mu \cdot \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix}$$

10.

- (a) Show that the curve represented by the parametric equations;

$$x = 2 + 4 \cos \theta \text{ and } y = 3 + 4 \sin \theta \text{ where } \theta \text{ is the parameter is a circle.} \quad [4]$$

- (b) Hence state the coordinates of the centre and the radius of the circle. [2]

- (c) Find the equation of the tangent to the curve where $\theta = \frac{1}{4}$ in the form $y = mx + c$,
where m and c are exact. [8]

- (d) Find the coordinates of the point where the tangent in (c) above meets
the x - axis. [2]

SOLUTION

- (a) $x = 2 + 4 \cos \theta$ and $y = 3 + 4 \sin \theta$

Using the identity;

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{x-2}{4}\right)^2 + \left(\frac{y-3}{4}\right)^2 = 1$$

$$\frac{(x-2)^2}{16} + \frac{(y-3)^2}{16} = 1$$

$$(x-2)^2 + (y-3)^2 = 16$$

$$(x-2)^2 + (y-3)^2 = 4^2 \text{ [shown]}$$

(b)

(i) Centre = (2; 3) and radius, $r = 4$ units

(c)

$$\frac{dx}{d\theta} = -4 \sin \theta \quad \text{and} \quad \frac{dy}{d\theta} = 4 \cos \theta$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{d\theta} \times \frac{d\theta}{dx} \\ &= \frac{4 \cos \theta}{-4 \sin \theta} \\ &= -\cot \theta \end{aligned}$$

$$\text{If } \theta = \frac{\pi}{4}; \text{ Gradient, } m \text{ of } T = -\frac{1}{\tan \frac{\pi}{4}} = -1$$

$$x^1 = 2 + 4 \cos \frac{\pi}{4}$$

$$= 2 + 2\sqrt{2}$$

$$y^1 = 3 + 4 \sin \frac{\pi}{4}$$

$$= 3 + 2\sqrt{2}$$

Therefore equation of the tangent;

$$y - y_1 = m(x - x_1)$$

$$y - 3 - 2\sqrt{2} = -1(x - 2 - 2\sqrt{2})$$

$$y = -x + 2 + 2\sqrt{2} + 3 + 2\sqrt{2}$$

$$y = -x + (5 + 4\sqrt{2}) \quad [\text{leave in the form } y = mx + c]$$

(d) At x -intercept; $y = 0$

$$x = 5 + 4\sqrt{2}$$

$$\therefore \text{Coordinates} = (5 + 4\sqrt{2}; 0)$$

11. The function $f(x) = x^2 + 2x$ for $x \in \mathbb{R}, -4 \leq x \leq -1$ (a) Express $f(x)$ in the form $(x + b)^2 + c$ where b and c are constants. [3]

(b) Find,

(i) $f^{-1}(x)$, [2]

(ii) the domain $f^{-1}(x)$. [1]

(c) Sketch on the same axis the graphs of $y = f^{-1}(x)$ and $f(x)$ showing clearly the relationship between the two graphs. [4]

SOLUTION

(a)

$$f(x) = x^2 + 2x$$

$$= x^2 + 2x + \left[\frac{1}{2}(2)\right]^2 - \left[\frac{1}{2}(2)\right]^2$$

$$= (x + 1)^2 - 1$$

$$a = 1; b = 1; c = -1$$

(b)

(i) Let $y = (x + 1)^2 - 1$

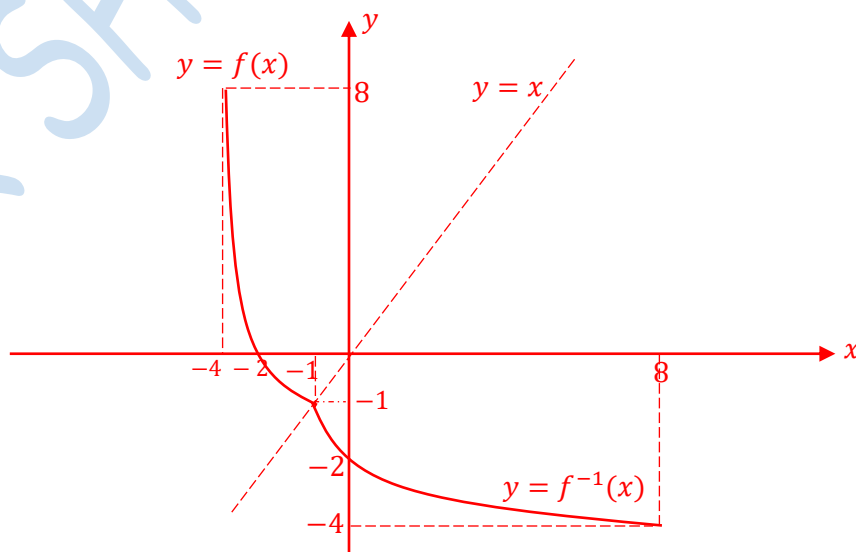
$$y + 1 = (x + 1)^2$$

$$x = -1 \pm \sqrt{y + 1}$$

$$\therefore f^{-1}(x) = -1 - \sqrt{x + 1}$$

(ii) Domain of $f^{-1}(x)$; $-1 \leq x \leq 8$ NB; Range of $f(x)$ is the domain of $f^{-1}(x)$

(c)



$y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$

(d) Let P_n be the given statement;

$$\text{when } n = 1; \quad LHS = 3(1) - 2 = 1 \quad \text{and} \quad RHS = \frac{1(3-1)}{2} = 1$$

therefore P_1 is true.

Assume P_n is true for $n = k$

$$1 + 4 + 7 + \dots + (3k - 2) = \frac{k}{2}(3k - 1)$$

To prove P_n is also true for $n = k + 1$;

$$\begin{aligned} 1 + 4 + 7 + \dots + (3(k + 1) - 2) &= \frac{k + 1}{2}(3(k + 1) - 1) \\ &= \frac{(k + 1)(3k + 2)}{2} \end{aligned}$$

LHS;

$$\begin{aligned} &= \frac{k(3k - 1)}{2} + 3k + 1 \\ &= \frac{3k^2 - k + 6k + 2}{2} \\ &= \frac{3k^2 + 5k + 2}{2} \\ &= \frac{3k(k + 1) + 2(k + 1)}{2} \\ &= \frac{(k + 1)(3k + 2)}{2} \equiv RHS \end{aligned}$$

$\therefore P_{k+1}$ is also true

By mathematical Induction P_n is true for $n = 1, n = k, n = k + 1$ and $\forall n \geq 1$

12.

(a) By sketching graphs of $y = x^2 - 4$ and $y = \frac{1}{x}$, determine the number of roots of the equation $x^2 - 4 = \frac{1}{x}$. [6]

(b) Show,

(i) by calculations that one of the roots of the equation $x^2 - 4 = \frac{1}{x}$ lies between 2 and 2.5. [4]

(ii) that the roots of the equation $x^2 - 4 = \frac{1}{x}$ satisfies the

$$\text{equation } x = \sqrt{\frac{1}{x} + 4}. \quad [2]$$

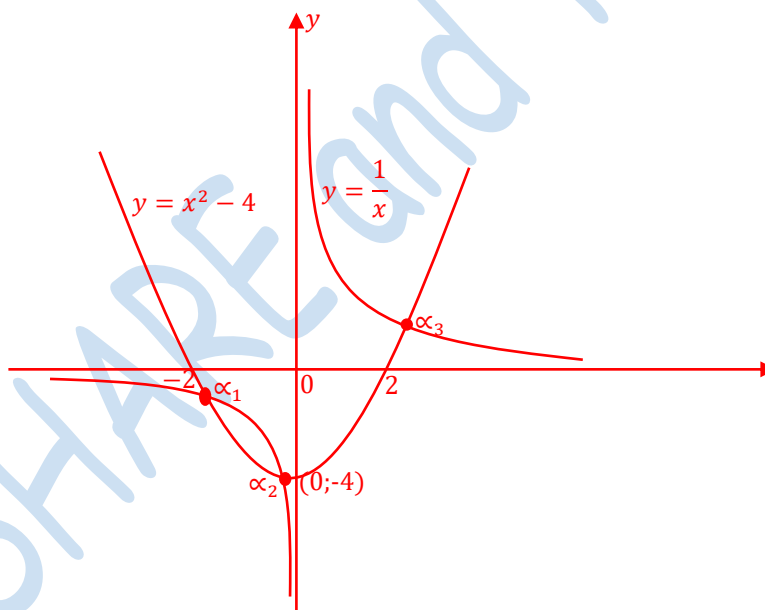
(c) Hence use the iterate $x_{n+1} = \sqrt{\frac{1}{x_n} + 4}$ taking $x_1 = 2.1$ as the first approximation to estimate the root of the equation to 3 decimal places. [4]

SOLUTION

(a) For $y = x^2 - 4$

If $x = 0$, $y = -4$ and If $y = 0$, $x = \pm 2$

For, $y = \frac{1}{x}$; asymptotes at $x = 0$ and $y = 0$



Hence 3 real roots since the two graphs intersect thrice.

(b)

(i) Let $f(x) = x^2 - 4 - \frac{1}{x}$

$$f(2) = 2^2 - 4 - \frac{1}{2} = -\frac{1}{2} < 0$$

$$f(2.5) = 2.5^2 - 4 - \frac{1}{2.5} = 1.85 > 0$$

Therefore change of sign indicates presence of root between 2 and 2.5

(ii) $x^2 - 4 = \frac{1}{x}$

$$x^2 = \frac{1}{x} + 4$$

$$x = \sqrt{\frac{1}{x} + 4} \text{ [shown]}$$

(c) $x_1 = 2.1$

$$x_2 = \sqrt{\frac{1}{x_1} + 4} = \sqrt{\frac{1}{2.1} + 4} = 2.115700942$$

$$x_3 = 2.114865621$$

$$x_4 = 2.114909757$$

$$x = 2.115 \text{ (3d.p)}$$

FEEL FREE TO CONTACT ME FOR ANY ADJUSTMENTS, CLARIFICATIONS
AND ASSISTANCE!

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"Concept before anything!", Author

Proverbs 11 vs. 2

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Tutors in:

- O - A Level Accounting
- O - A Level Mathematics
- O - A Level Additional/ Further Mathematics
- Business Mathematics
- Engineering Mathematics
- Actuarial Mathematics
- Quantitative Techniques and Methods

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